The separability lemma

Notation

- V — the rating variables; $t \subseteq V$ is a set of variables
- $M \subseteq \mathcal{P}(V)$ — the sets that carry a term in the model
- $g_t(x_t)$ — the term on t; $\beta_t = \ln g_t$ is its coefficient table
- $T \in M$ — the term whose table we want; $\bar{T} = V \setminus T$
- x — a row; z_T — chosen values for the variables in T; z^0 — base values
- $(z_T;\, x_{\bar{T}})$ — the row with T fixed at z and everything else left as it is
- $E_P[\,\cdot\,]$ — exposure-weighted mean over the rows of a data set P

Three relations to T

Every set of variables $t \subseteq V$ stands in exactly one relation to T:

$$\begin{aligned} & t \subseteq T && \text{inside } T \\ t \cap T = \emptyset, \quad & t \neq \emptyset && \text{disjoint from } T \\ t \cap T \neq \emptyset, \quad & t \not\subseteq T && \text{overlapping } T \text{ partway} \end{aligned}$$

This partitions $\mathcal{P}(V)$. The model's terms inherit it:

$$Q = M \cap \mathcal{P}(T), \qquad Q' = \{\, t \in M : t \cap T = \emptyset,\ t \neq \emptyset \,\}, \qquad Q'' = M \setminus (Q \cup Q')$$

Example. Terms age, sex, breed, weight, deductible, age×weight, breed×weight; $T = \{\text{age}\}$. $Q = \{\emptyset, \text{age}\}$, $Q' = \{\text{sex}, \text{breed}, \text{weight}, \text{deductible}, \text{breed}\times\text{weight}\}$, $Q'' = \{\text{age}\times\text{weight}\}$.

The model, term by term

$$\ln \hat{\mu}(x) = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\sum_{|t|=1} \beta_t(x_t)}_{\text{main effects}} + \underbrace{\sum_{|t|\geq 2} \beta_t(x_t)}_{\text{interactions}}$$

As a product:

$$\hat{\mu}(x) = \prod_{t \in M} g_t(x_t), \qquad g_t = e^{\beta_t}, \qquad g_\emptyset = e^{\beta_0}$$

Each factor sees only its own variables.

Fix T, sort the terms by overlap

Evaluate at $(z_T;\, x_{\bar{T}})$. A term in Q sees only fixed values; a term in Q' sees only the row's values; a term in Q'' sees both:

$$\hat{\mu}(z_T;\, x_{\bar{T}}) = \underbrace{\prod_{t \in Q} g_t(z_t)}_{G_Q(z)} \cdot \underbrace{\prod_{t \in Q'} g_t(x_t)}_{G_{Q'}(x)} \cdot \underbrace{\prod_{t \in Q''} g_t(z_{t \cap T};\, x_{t \setminus T})}_{G_{Q''}(z,x)}$$

Only $G_{Q''}$ mixes z and x.

If $Q'' = \emptyset$: average over any P, take the ratio to base, and $G_{Q'}$ cancels — P with it.

What Q'' depends on, and how to pin it

$$C = \bigcup_{t \in Q''} (t \setminus T) \qquad \text{the variables } Q'' \text{ reaches outside } T$$

Hold them at base, $x_C = z^0_C$:

$$\hat{\mu}(z_T;\, z^0_C;\, x_{\overline{T \cup C}}) = \underbrace{\prod_{t \in Q} g_t(z_t) \cdot \prod_{t \in Q''} g_t(z_{t \cap T};\, z^0_{t \setminus T})}_{z \text{ only}} \cdot \underbrace{\prod_{t \in Q'} g_t(z^0_{t \cap C};\, x_{t \setminus C})}_{x \text{ only}}$$

- Every Q'' term is now a function of z alone; under treatment contrasts each equals 1
- Q' terms were never functions of z, and still aren't
- Average over any P, take the ratio to base: the x-only product cancels, and P is gone

Example: A, B, C, AB, BC

$$\ln \hat{\mu}(a,b,c) = \beta_0 + \beta_A(a) + \beta_B(b) + \beta_C(c) + \beta_{AB}(a,b) + \beta_{BC}(b,c)$$

$$\hat{\mu}(a,b,c) = g_\emptyset \, g_A(a) \, g_B(b) \, g_C(c) \, g_{AB}(a,b) \, g_{BC}(b,c)$$

Fix T; the variables not in T keep their row values x_A, x_B, x_C:

$$T = \{A\}: \quad \hat{\mu}(a, x_B, x_C) = \underbrace{g_\emptyset \, g_A(a)}_{Q} \cdot \underbrace{g_B(x_B) \, g_C(x_C) \, g_{BC}(x_B, x_C)}_{Q'} \cdot \underbrace{g_{AB}(a, x_B)}_{Q''}$$

$$T = \{B\}: \quad \hat{\mu}(x_A, b, x_C) = \underbrace{g_\emptyset \, g_B(b)}_{Q} \cdot \underbrace{g_A(x_A) \, g_C(x_C)}_{Q'} \cdot \underbrace{g_{AB}(x_A, b) \, g_{BC}(b, x_C)}_{Q''}$$

$$T = \{C\}: \quad \hat{\mu}(x_A, x_B, c) = \underbrace{g_\emptyset \, g_C(c)}_{Q} \cdot \underbrace{g_A(x_A) \, g_B(x_B) \, g_{AB}(x_A, x_B)}_{Q'} \cdot \underbrace{g_{BC}(x_B, c)}_{Q''}$$

$$T = \{A, B\}: \quad \hat{\mu}(a, b, x_C) = \underbrace{g_\emptyset \, g_A(a) \, g_B(b) \, g_{AB}(a,b)}_{Q} \cdot \underbrace{g_C(x_C)}_{Q'} \cdot \underbrace{g_{BC}(b, x_C)}_{Q''}$$

$$T = \{B, C\}: \quad \hat{\mu}(x_A, b, c) = \underbrace{g_\emptyset \, g_B(b) \, g_C(c) \, g_{BC}(b,c)}_{Q} \cdot \underbrace{g_A(x_A)}_{Q'} \cdot \underbrace{g_{AB}(x_A, b)}_{Q''}$$

Recovering the coefficients

Pin the variables that Q'' reaches outside T at base, average over the rest, take the ratio to base. Treatment contrasts throughout.

Main effects:

$$e^{\beta_A(a)} = \frac{E_P[\hat{\mu}(a, b_0, x_C)]}{E_P[\hat{\mu}(a_0, b_0, x_C)]} \quad \text{pin } b$$

$$e^{\beta_B(b)} = \frac{\hat{\mu}(a_0, b, c_0)}{\hat{\mu}(a_0, b_0, c_0)} \qquad \text{pin } a \text{ and } c \text{ — nothing left to average}$$

$$e^{\beta_C(c)} = \frac{E_P[\hat{\mu}(x_A, b_0, c)]}{E_P[\hat{\mu}(x_A, b_0, c_0)]} \quad \text{pin } b$$

Interactions: the pinned joint table, then the alternating product over its corners:

$$R_{AB}(a,b) = \frac{\hat{\mu}(a,b,c_0)}{\hat{\mu}(a_0,b_0,c_0)}, \quad e^{\beta_{AB}(a,b)} = \frac{R_{AB}(a,b)\; R_{AB}(a_0,b_0)}{R_{AB}(a,b_0)\; R_{AB}(a_0,b)} \quad \text{pin } c$$

$$R_{BC}(b,c) = \frac{\hat{\mu}(a_0,b,c)}{\hat{\mu}(a_0,b_0,c_0)}, \quad e^{\beta_{BC}(b,c)} = \frac{R_{BC}(b,c)\; R_{BC}(b_0,c_0)}{R_{BC}(b,c_0)\; R_{BC}(b_0,c)} \quad \text{pin } a$$

The same, on the log scale

Main effects: one plus, one minus.

$$\begin{aligned}
\beta_A(a) &= \ln E_P[\hat{\mu}(a,b_0,x_C)] - \ln E_P[\hat{\mu}(a_0,b_0,x_C)] \\
\beta_B(b) &= \ln \hat{\mu}(a_0,b,c_0) - \ln \hat{\mu}(a_0,b_0,c_0) \\
\beta_C(c) &= \ln E_P[\hat{\mu}(x_A,b_0,c)] - \ln E_P[\hat{\mu}(x_A,b_0,c_0)]
\end{aligned}$$

Interactions: sign $(-1)^{\text{number of coordinates at base}}$.

$$\begin{aligned}
\beta_{AB}(a,b) &= \ln R_{AB}(a,b) - \ln R_{AB}(a,b_0) - \ln R_{AB}(a_0,b) + \ln R_{AB}(a_0,b_0) \\
&= \ln \hat{\mu}(a,b,c_0) - \ln \hat{\mu}(a,b_0,c_0) - \ln \hat{\mu}(a_0,b,c_0) + \ln \hat{\mu}(a_0,b_0,c_0) \\
\beta_{BC}(b,c) &= \ln R_{BC}(b,c) - \ln R_{BC}(b,c_0) - \ln R_{BC}(b_0,c) + \ln R_{BC}(b_0,c_0) \\
&= \ln \hat{\mu}(a_0,b,c) - \ln \hat{\mu}(a_0,b,c_0) - \ln \hat{\mu}(a_0,b_0,c) + \ln \hat{\mu}(a_0,b_0,c_0)
\end{aligned}$$

From a GLM with treatment contrasts

The fit has done all of it.

Treatment coding makes $g_t = 1$ whenever any coordinate of t is at base. So at the base cell every Q'' term is already 1, for every T, and

$$\text{table}_t(z_t) = e^{\beta_t(z_t)} \qquad \text{for every } t \in M$$

Read the tables off the coefficients. No data, no averaging, no choice of P.

What the lemma adds:

- The same tables come back from the recipe on the next slide applied to `predict()`. The coefficients *are* that recipe, done in advance by the parameterization.
- A partial-dependence table for a variable agrees with e^{β} only when the variable sits in no interaction ($Q'' = \emptyset$). Otherwise the two differ, on the training data too, and the coefficient is the rating table.

- Other contrasts (sum-to-zero, Helmert) give the same $\hat{\mu}$ in other coordinates. Refit with treatment contrasts, or use the recipe.

From a GAM, a penalized fit, random effects

Same term list, but the fit chose the gauge.

`s(age) + s(weight) + ti(age, weight) + s(breed, bs = "re")` has a term list M. But the smooths are centered on the training data, `ti()` is centered on the training data, random effects are centered at zero. Nothing is 1 at a base level, and `predict(type = "terms")` returns these centered pieces: for a T with $Q'' \neq \emptyset$ (age, because of `ti(age, weight)`) the centered main effect carries a training-book average of the interaction.

Recipe — only `predict()`, never the coefficients:

1. Choose a base cell z^0: a well-populated level of each variable. Random effects have no reference level, so this choice is yours.
2. For each $t \in M$, hold everything outside t at base, vary t's variables, and take the alternating product over the corners of t:

$$\text{table}_t(z_t) = \prod_{u \subseteq t} \hat{\mu}\big(z^{(u)}\big)^{(-1)^{|t \setminus u|}}, \qquad z^{(u)} = \text{the base cell with the variables in } u \text{ moved to } z$$

$|t| = 1$: a ratio. $|t| = 2$: the double ratio of slide 7. A smooth gives a curve: evaluate on a grid of z.

Upstream of the table: a penalty is not invariant to reparameterization, so the fit's gauge also shapes $\hat{\mu}$ itself. That is a modeling choice; the recipe takes $\hat{\mu}$ as given.

From a black box

No term list. Assume every Q'' reaches every variable outside T.

Then the lemma says: no averaged table is mix-free. Partial dependence, ALE, model one-ways all move with the book, for every variable. The only mix-free object is the profile through the base cell.

Recipe — the same formula, with nothing optional about the pinning:

1. Choose the base cell z^0 deliberately. Every table depends on where every other variable sits, so use a well-populated cell and report it with the tables.
2. Choose which tables the rating plan carries: main effects and the pairs you want.
3. For each:

$$\text{table}_t(z_t) = \prod_{u \subseteq t} \hat{\mu}\big(z^{(u)}\big)^{(-1)^{|t \setminus u|}}$$

4. Check the truncation. With all $2^{|V|}$ tables the product reproduces $\hat{\mu}$ exactly; the tables you leave out are the residual. Compare $\prod_{t \text{ kept}} \text{table}_t(x_t)$ with $\hat{\mu}(x)$ on the book.